Design and Process Study of Autoclave-Cured Prepreg Composites

Study ideation and working of the decision tool (version 5.5)

1. Study Ideation

1.1 The problem

An autoclave-cured carbon/epoxy part is decided by three linked choices: the prepreg, the mold (tool) it is cured on, and the cure cycle. In practice these are usually picked from experience or habit, and the reasons are rarely written down. A wrong pair can cost tolerance: when the tool and the part expand by different amounts during heating, the cured part comes off the tool the wrong size or warped, and corners spring in. The study asks how far these choices, and the process settings, drive the dimensional accuracy of the cured part.

1.2 Problem statements

Code	Question	Where the tool answers it
PS1 — Tool-part thermal interaction	How do CTE mismatch and spring-in affect the geometric tolerance of the cured part?	Tooling tab; DOE factor *tool material*
PS2 — Cure-cycle sensitivity	How do pressure, ramp rate and dwell time relate to dimensional deviation?	Cure tab; DOE factors *pressure, ramp, dwell*
PS3 — Layup and geometry effects	How do stacking sequence and geometry drive warpage and tolerance stack-up?	DOE factors *layup, thickness, corner geometry*

1.3 How the idea developed

- **Starting point.** The team's composite material recommender (versions 1-6) ranked fibers, prepregs and resins with a rule-based weighted score.
- **Defensible weights.** Hand-set weights were replaced by AHP (Analytic Hierarchy Process): criteria are compared in pairs, weights come from the eigenvector, and a consistency ratio below 0.10 shows the judgements are not contradictory.
- **Sourced data.** Datasheet and handbook values (Hexcel, Toray, NCAMP, MIL-HDBK-17) were added, with the source of every number recorded.
- **From a recommender to a study tool.** The project documents (Master Report, Matrix Engine Report, BOM and cost sheet, HexPly M18/1 guide, ANSYS thermal calculation, spring-in summary) turned the recommender into a tool that also selects the mold, plans the cure, builds the test matrix and prices the study.
- **Requirement-driven start (v5.4).** Instead of assuming a fixed 1 m² M18/1 sandwich panel, the study now starts from what the laminate must achieve; the prepreg and its datasheet cure cycle follow from that.

1.4 Design principles

- **Every number has a source.** Values are typed in the source's own units and converted in code; anything assumed is flagged as an assumption.
- **Missing data is never guessed.** A property that is not published scores as "not demonstrated" (zero), and 40 prepreg systems without laminate data are listed as "datasheet needed" rather than given invented values.
- **Every rejection is explained.** The user sees why each prepreg or tool failed, for example "strength 2207 MPa < 2500 required".
- **Rank and estimate, not replace.** The tool narrows choices and predicts trends; it does not replace FEM distortion analysis (ANSYS ACP/ACCS), supplier qualification or trial parts.

1.5 Expected outcome of the study

A traceable chain from requirement to cured part: which prepreg meets the requirement, which tool keeps the part in tolerance, which cure cycle to run and how long it takes, which factors to test (and in what run order), what the study costs, and, once parts are measured, which factors actually moved the dimensions.

2. Working of the Project

The tool is a desktop program (*run.bat* opens *project_gui.py*; a text-menu version is *app.py*). Its tabs follow the study from left to right, and each tab passes its result to the next one.

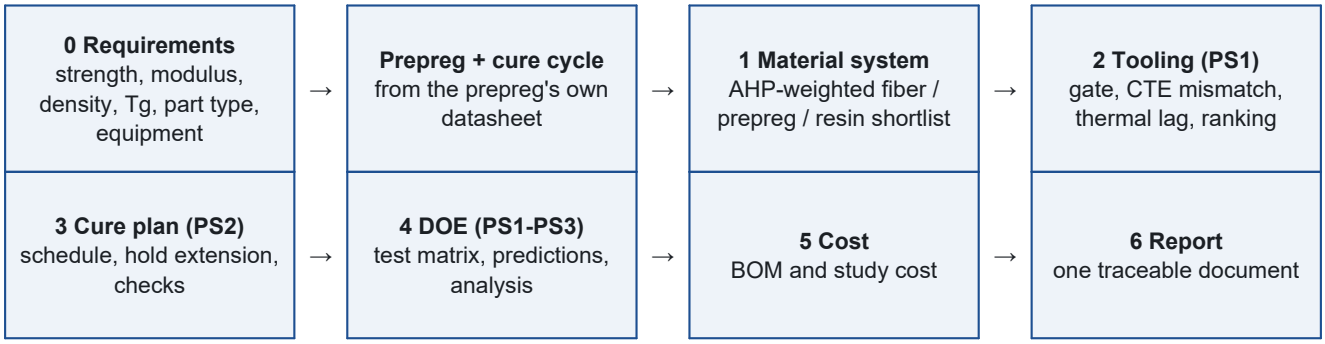


Figure 1 — Workflow of the decision tool.

Step 0 — Requirements to prepreg and cure cycle

The user enters the minimum 0° tensile strength (MPa), and optionally the minimum modulus (GPa), maximum laminate density (g/cm³), minimum dry Tg (°C) and fiber form, plus the part size, part type (monolithic or honeycomb sandwich), skin thickness and equipment (autoclave or oven).

- **Data.** Prepregs come from the project prepreg database (48 systems). Cured-laminate values come from each system's datasheet: 13 fiber versions of 7 systems (HexPly 8552, M20, M21, M56, M18/1, Toray 3900).
- **Pass / fail filters.** Strength, modulus, density, Tg, fiber form, and whether the datasheet gives a cure cycle for this part type and equipment.
- **Ranking.** Survivors are scored 0-1: specific strength (strength ÷ density) 50 %, specific modulus 30 %, Tg 20 %.
- **Cure cycle.** Taken from the winner's own datasheet: oven → vacuum-bag-only cycle; sandwich → sandwich cycle; over 15 mm thick → thick-part cycle (M21); otherwise the monolithic cycle. Its temperature, pressure and ramp become the study values for every later tab.

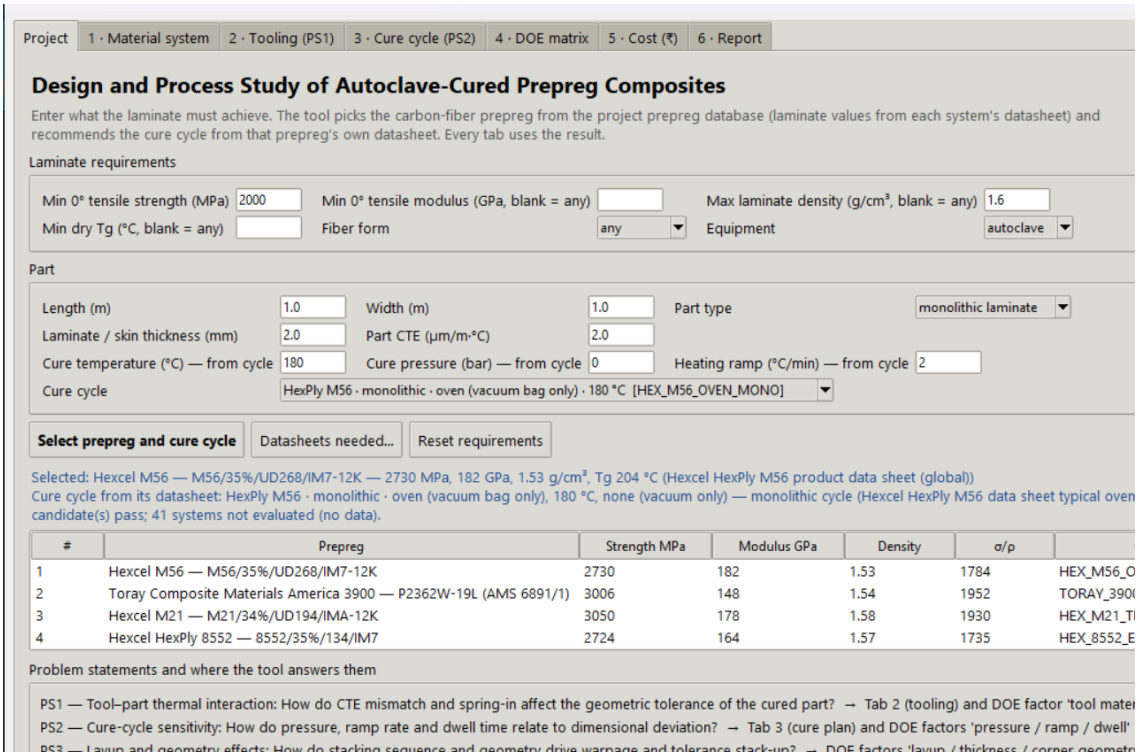


Figure 2 — Project tab: requirements in, ranked prepregs and the datasheet cure cycle out.

Step 1 — Material system

The fiber / prepreg / resin recommender ranks the databases with an AHP-weighted score. Each property is min-max normalised (best = 1, worst = 0), multiplied by its AHP weight and summed. Ties and close calls are marked, and a lamina estimator predicts UD properties from fiber data (fitted on 27 published fiber-lamina pairs; it refuses to extrapolate).

Step 2 — Tooling (PS1)

- **Gate.** Each of 9 tool materials must survive the cure (service temperature ≥ cure + 20 °C), the number of cycles, the cost ceiling, the tolerance, the surface finish and the autoclave size; every rejection is explained.
- **CTE mismatch error.** $\delta = |\alpha_{tool} - \alpha_{part}| \times \Delta T \times L$ — how much the tool and part grow differently over the temperature rise ΔT and part length L. Invar uses its temperature-dependent CTE.
- **Thermal lag.** $\tau = \rho\, c_p\, t / h$ — how slowly the tool heats, using measured autoclave heat-transfer coefficients h (Fisher et al.: about 37 W/m²K oven, 99 at 3 bar, 138 at 7 bar).
- **Ranking.** Error, lag, mass, cycle life and cost (INR) are normalised and weighted by an AHP profile (prototype, production or aerospace). A ±20 % weight sweep flags close calls.

Step 3 — Cure plan (PS2)

- **Library.** 33 documented cycles (Hexcel, Toray, NCAMP, MIL-HDBK-17, project files) with ramp-rate and hold-time ranges, staged pressure and vacuum, vent temperatures and post-cures.
- **Schedule.** Temperature, pressure and vacuum against time, with fast / nominal / slow cycle-time bands.
- **Tool lag correction.** The final hold is extended by $\tau \times \ln(\text{lag} / 5\ °C)$ so the part, not just the air, reaches temperature.
- **Checks.** Sandwich over-pressure, oven vs autoclave cycle, thickness vs a thin-part cycle, exotherm dwell and tool rating. An ANSYS input table is exported.

Step 4 — Design of experiments (PS1-PS3)

Seven two-level factors are available: tool material, pressure (4 / 6 bar), ramp (1 / 3 °C/min), dwell (120 / 180 min), layup (QI symmetric / unbalanced), skin thickness (2 / 4 mm) and corner geometry (flat / 90°). The scoping question pre-selects factors; the tool builds a full or half factorial matrix with replicates in random run order (24 runs by default) and predicts δ, cycle time and spring-in for each run (Albert & Fernlund: about 0.90° UD, 1.28° QI at a 90° corner). After CMM / 3D-scan measurement, the main effect of each factor and its t-ratio show which factors really moved the dimensions.

Step 5 — Cost and Step 6 — Report

The cost tab builds the panel bill of materials (prepreg; honeycomb core and film adhesive for sandwich parts; peel ply, release film, bleeder, breather, bagging film, sealant tape, vacuum fittings and thermocouples) and the study cost in INR from the project BOM sheet; values not confirmed by a quote are marked unverified. The report tab joins every section — prepreg selection, material, tooling, cure plan, DOE, analysis and cost — into one document.

Verification

Check	Result
Unit tests (engines, cure library, prepreg selector, desktop window)	82 / 82 pass
Validation cases — reproduce accepted industry choices	6 / 6 reproduce the documented choice
Prepreg selector	Every candidate meets every requirement; impossible requests return none with the closest rejections
End-to-end run	Requirements → prepreg → tooling → cure → 24-run DOE → report, no errors

Limitations

- Datasheet values are typical values, not design allowables; NCAMP-measured means for 8552 run 7-9 % below the datasheet.
- 40 prepreg systems need datasheets before they can be selected by strength.
- Predictions use closed-form physics; final distortion needs FEM and trial parts.